



Date: 27 / 9 / 2026

Lab Practical #9

Define and design a network infrastructure of any organization.

Practical Assignment #9:

A. Define and design a network infrastructure that includes the following components:

- Web Server - can use HTTP protocol
- Data Server - can use ftp protocol
- Email Server - can use mail protocol
- Local Web Server
- Three different networks - can use router
- Five individual LAN segments - can use VLAN
- One Wi-Fi router - can create small wireless network.

To do Instructions:

- 1. Required followed things: All Topology in One Topology**

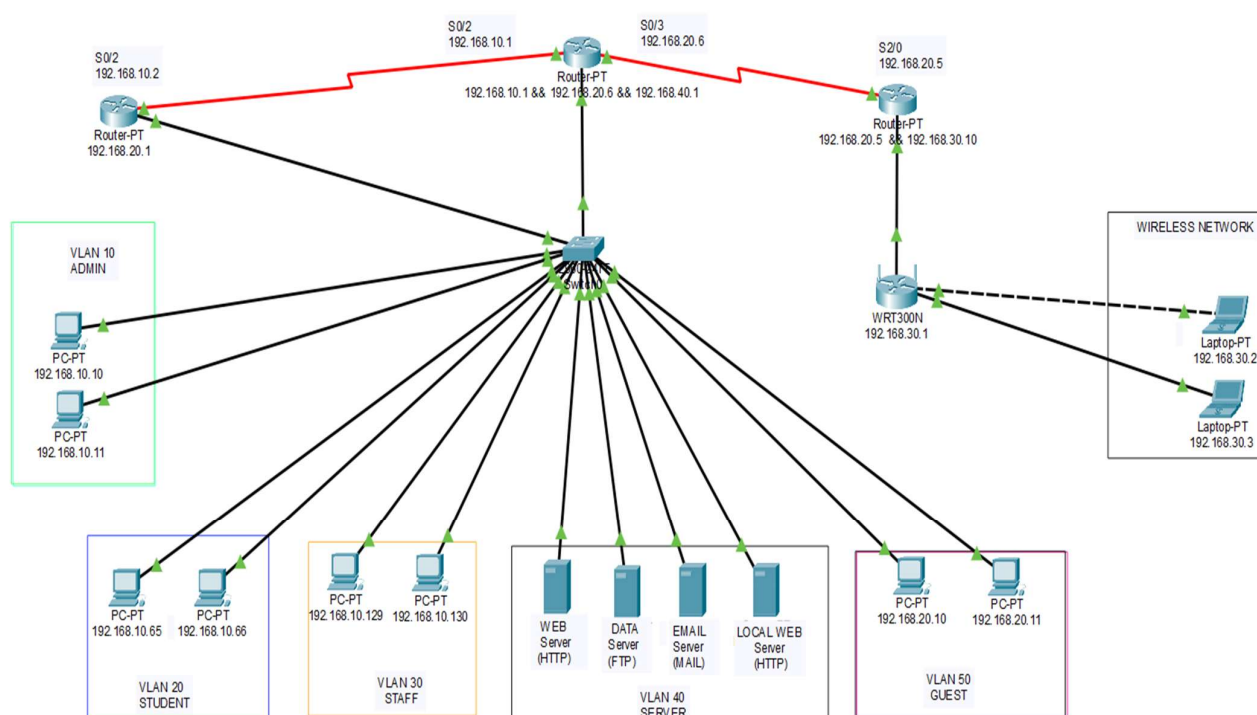
1. Define:

A **network infrastructure** is a collection of hardware devices, servers, communication links, and networking protocols that work together to allow computers and other devices to communicate and share resources.

The proposed network infrastructure contains **three different networks, five VLAN-based LAN segments, four types of servers, and one wireless network**. Routers are used to connect different networks, while VLANs are used to divide the LAN into separate logical segments.



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DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY
Semester 5th | Practical Assignment | Computer Networks (2301CS501)

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Sr. No.	Device	Interface / VLAN	IP Address	Service / Purpose
1	Router 1	S0/2	192.168.10.2	Network 1
2	Router 1	LAN	192.168.20.1	Network 2
3	Router 2(Left)	S0/2	192.168.10.1	Network 1
4	Router 2(Right)	S0/3	192.168.20.6	Network 2
5	Router 2	LAN	192.168.40.1	Network 2
6	Router 3	S2/0	192.168.20.5	Network 2
7	Router 3	LAN	192.168.30.10	Network 3
8	Wi-Fi Router	LAN	192.168.30.1	Wireless Gateway
9	PC-1	VLAN 10	192.168.10.10	Admin
10	PC-2	VLAN 10	192.168.10.11	Admin
11	PC-3	VLAN 20	192.168.10.65	Student
12	PC-4	VLAN 20	192.168.10.66	Student
13	PC-5	VLAN 30	192.168.10.129	Staff
14	PC-6	VLAN 30	192.168.10.130	Staff
15	Web Server	VLAN 40	192.168.10.194	HTTP
16	Data Server	VLAN 40	192.168.10.195	FTP
17	Email Server	VLAN 40	192.168.10.196	SMTP/POP3
18	Local Web Server	VLAN 40	192.168.10.197	HTTP
19	PC-7	VLAN 50	192.168.20.10	Guest
20	PC-8	VLAN 50	192.168.20.11	Guest
21	Laptop-1	Wi-Fi	192.168.30.2	Wireless
22	Laptop-2	Wi-Fi	192.168.30.3	Wireless